Worksheet 4.1 — Increasing/Decreasing & the First Derivative Test

Grade 12 Calculus and Vectors (MCV4U) · Unit 4: Curve Sketching

The Big Ideas

f increases where $f' > 0$ and decreases where $f' < 0$. Critical numbers ($f' = 0$ or undefined) are classified by how f' changes sign.

- Increasing: $f' > 0$; decreasing: $f' < 0$.
- Critical numbers: $f' = 0$ or undefined.
- 1st-derivative test: $+ \to -$ max, $- \to +$ min, no change neither.

The Big Picture

Sign of the derivative

Where $f'(x) > 0$ the function **increases**; where $f'(x) < 0$ it **decreases**. The derivative's sign reads the graph's direction.

Critical numbers

Points where $f'(x) = 0$ or is undefined are **critical numbers** the only places a local max or min can occur.

The first derivative test

If f' changes from $+$ to $-$ at a critical number it is a local maximum; from $-$ to $+$ it is a local minimum.

Worked Examples

Example 1 — Intervals

Where is $f(x) = x^2 - 4x$ increasing/decreasing?

Solution. **Step 1 (differentiate):** $f'(x) = 2x - 4$.

Step 2 (find where $f' = 0$, the boundary between the intervals): $2x - 4 = 0 \Rightarrow x = 2$.

Step 3 (test the sign of f' on each side): for $x < 2$, e.g. $x = 0$: $f'(0) = -4 < 0$ (decreasing). For $x > 2$, e.g. $x = 3$: $f'(3) = 2 > 0$ (increasing).

Conclusion: decreasing on $(-\infty, 2)$, increasing on $(2, \infty)$.

Example 2 — Critical numbers

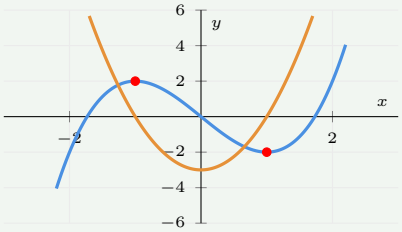
Critical numbers of $f(x) = x^3 - 3x$.

Solution. **Step 1 (differentiate):** $f'(x) = 3x^2 - 3$.

Step 2 (set $f' = 0$): $3x^2 - 3 = 0 \Rightarrow x^2 = 1$.

Step 3 (solve): $x = \pm 1$.

Conclusion: critical numbers at $x = -1$ and $x = 1$. f rises where $f' > 0$, falls where $f' < 0$:



Example 3 — Classify

Classify the critical points of $f(x) = x^3 - 3x$.

Solution. **Step 1 (recall the critical numbers):** from the previous example, $x = -1$ and $x = 1$, with $f'(x) = 3x^2 - 3$.

Step 2 (test the sign of f' around $x = -1$): f' goes from $+$ (e.g. $f'(-2) = 9 > 0$) to $-$ (e.g. $f'(0) = -3 < 0$) — a local max.

Step 3 (test the sign of f' around $x = 1$): f' goes from $-$ to $+$ (e.g. $f'(2) = 9 > 0$) — a local min.

Conclusion: local max at $(-1, 2)$; local min at $(1, -2)$.

Example 4 — Neither

Classify the critical point of $f(x) = x^3$.

Solution. **Step 1 (differentiate and set to 0):** $f'(x) = 3x^2 = 0 \Rightarrow x = 0$.

Step 2 (test the sign of f' on each side): $f'(x) = 3x^2 \geq 0$ for every x — it's $+$ on both sides of 0, never negative.

Step 3 (interpret): no sign change means neither a max nor a min.

Conclusion: $x = 0$ is neither (it's an inflection point instead).

Example 5 — Several

Find and classify the critical points of $f(x) = x^4 - 2x^2$.

Solution. **Step 1 (differentiate):** $f'(x) = 4x^3 - 4x$.

Step 2 (factor and set to 0): $4x(x^2 - 1) = 4x(x - 1)(x + 1) = 0 \Rightarrow x = 0, 1, -1$.

Step 3 (test the sign of f' in each of the four intervals): the signs go $-, +, -, +$ moving left to right through $x = -1, 0, 1$.

Conclusion: local min at $x = -1$ ($f(-1) = -1$), local max at $x = 0$ ($f(0) = 0$), local min at $x = 1$ ($f(1) = -1$).

Example 6 — Intervals

Where is $f(x) = x^2 - 6x$ decreasing?

Solution. **Step 1 (differentiate):** $f'(x) = 2x - 6$.

Step 2 (set up the decreasing condition, $f' < 0$): $2x - 6 < 0$.

Step 3 (solve the inequality): $2x < 6 \Rightarrow x < 3$.

Conclusion: decreasing on $(-\infty, 3)$.

Example 7 — Critical numbers

Critical numbers of $f(x) = x^3 - 12x$.

Solution. **Step 1 (differentiate):** $f'(x) = 3x^2 - 12$.

Step 2 (set $f' = 0$ and solve): $3x^2 = 12 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$.

Conclusion: critical numbers at $x = -2$ and $x = 2$.

Example 8 — Always increasing

Is $f(x) = x^3 + x$ ever decreasing?

Solution. **Step 1 (differentiate):** $f'(x) = 3x^2 + 1$.

Step 2 (analyze the sign): $3x^2 \geq 0$ for every x, so $3x^2 + 1 \geq 1 > 0$ always.

Conclusion: $f'(x) > 0$ for every x, so f is never decreasing.

Example 9 — Classify

Classify the critical point of $f(x) = x^2 - 4x$.

Solution. **Step 1 (differentiate and find the critical number):** $f'(x) = 2x - 4 = 0 \Rightarrow x = 2$.

Step 2 (test the sign on each side): for $x < 2$, $f' < 0$ (decreasing); for $x > 2$, $f' > 0$ (increasing) — a $- \to +$ pattern.

Conclusion: local min at $x = 2$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Where is $f(x) = x^2 - 6x$ increasing?

Question 2

Critical numbers of $f(x) = x^3 - 12x$?

Question 3

Classify the critical points of $f(x) = x^2 - 4x$.

Question 4

Is $f(x) = x^3 + x$ ever decreasing?

Question 5

Critical numbers of $f(x) = x^4 - 2x^2$?

Question 6

Where is $f(x) = x^2 + 2x$ decreasing?

Question 7

Critical numbers of $f(x) = x^3 - 3x^2$?

Question 8

At a critical number f' goes $- \to +$. What is it?

Question 9

Classify the critical point of $f(x) = x^3$.

Question 10

Where is $f(x) = x^3 - 3x$ increasing?

Question 11

Is $f(x) = -x^2$ increasing or decreasing for $x > 0$?

Question 12

Local max value of $f(x) = x^3 - 3x$?

Question 13 — Challenge

Find and classify all critical points of $f(x) = x^3 - 6x^2 + 9x$.

Question Recap & Answer Key

Question Recap

1. Where is $f(x) = x^2 - 6x$ increasing?
2. Critical numbers of $f(x) = x^3 - 12x$?
3. Classify the critical points of $f(x) = x^2 - 4x$.
4. Is $f(x) = x^3 + x$ ever decreasing?
5. Critical numbers of $f(x) = x^4 - 2x^2$?
6. Where is $f(x) = x^2 + 2x$ decreasing?
7. Critical numbers of $f(x) = x^3 - 3x^2$?
8. At a critical number f' goes $- \to +$. What is it?
9. Classify the critical point of $f(x) = x^3$.
10. Where is $f(x) = x^3 - 3x$ increasing?
11. Is $f(x) = -x^2$ increasing or decreasing for $x > 0$?
12. Local max value of $f(x) = x^3 - 3x$?
13. Find and classify all critical points of $f(x) = x^3 - 6x^2 + 9x$.

✓ Answer Key

1. $x > 3$
2. $x = \pm 2$
3. local min at $x = 2$
4. no ($f' = 3x^2 + 1 > 0$)
5. $0, \pm 1$
6. $x < -1$
7. $0, 2$
8. local min
9. neither (inflection)
10. $|x| > 1$
11. decreasing
12. 2 (at $x = -1$)
13. $f' = 3x^2 - 12x + 9 = 3(x-1)(x-3)$: max $(1, 4)$, min $(3, 0)$